GG–M–117
DECEMBER 19, 1955
SUPERSEDING
Int. Fed. Spec. GG–M–00117 (DOD–SHIPS)
1 October 1954

FEDERAL SPECIFICATION

MARKER, FOUNTAIN TYPE, FELT TIP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers felt-tip fountain-type markers (known commercially as felt-tip marking pens) for use in freehand marking, writing, drawing, and identification marking of various types of material.

1.1.1 *Scope of illustrations.*—THE ILLUSTRATIONS SHOWN HEREIN ARE DESCRIPTIVE AND NOT RESTRICTIVE and are included for the convenience of requisitioning and purchasing officers and manufacturers, and are not intended to preclude the purchase of markers which are otherwise in accordance with the requirements of this specification.

1.2 Classification.

1.2.1 *Types, classes, and styles.* — Markers shall be of the following types, classes, and styles, as specified (see 6.1):

Type I.—Valve control feed.
  Class 1.—Small (pocket size).
  Class 2.—Large (heavy duty).

Type II.—Adjustable control feed.

Type III.—Gravity feed.
  Class 1.—Small (pocket size).
    Style A.—Metal.
    Style B.—Composition.
    Style C.—Self-filling.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. — The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Military Specifications:

MIL–R–2765—Rubber Material, Synthetic, Oil-Resistant (Sheet, Strip, and Molded Shapes).

MIL–P–15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of

362246—55—No. 135

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military specifications, standards, drawings, and other publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.—The following publications, of the issues in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

Department of Commerce Publication:

Commercial Standard 185–52—Wool Felt.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

Consolidated Classification Committee Publication:

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Material.—The material shall be as hereinafter specified.

3.2 Design.—The markers shall be designed to operate satisfactorily with a quick-drying noncorrosive ink; either the thin, fast-flowing dye ink or the heavier bodied, slower flowing, pigmented or metallic inks.

3.3 Felt nib.—The felt nib shall be firmly maintained in the writing head and be readily replaceable.

3.3.1 *Felt.* — The felt nib shall be a single unit of either 26S1 (hard) or 32S1 (extra hard) wool felt for use with thin dye ink and not less than 20S1 (medium) wool felt for use with metallic or pigmented ink, as specified (see 6.1). The felt shall be in accordance with Commercial Standard 185–52.

3.3.2 *Shape of felt nib.* — Felt nibs shall be either round, conical, chisel, square, or T-shaped, as specified (see 6.1).

3.3.3 *Quantity of felt nibs.*—When specified, additional felt nibs shall be furnished with each marker (see 6.1).

3.4 Finish. — Markers shall have a smooth, even surface, thoroughly cleaned and substantially free from scale, pits, burrs, cracks, sharp projections, and other detrimental defects. The external surfaces of the markers shall be either buff polished, plated, enameled, or anodized. Surfaces not usually ground or buffed shall have a surface finish conforming to good manufacturing practice.

3.5 Identification marking.—Markers shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture can be readily determined.

3.6 Type I, valve control feed. — Type I markers shall be provided with a spring-actuated leakproof valve to control and govern the ink flow, and shall consist essentially of a body or barrel, writing head, cap, and felt nib.

3.6.1 *Body (barrel).*—The body shall be of metal and shall be provided with internal threads at the open end of the ink chamber for attaching and securing the writing head. A solvent-resistant plastic washer shall be provided between the bearing surfaces of the body and writing head to prevent ink leakage after the writing head is assembled to the body. The outside diameter of the body at the writing-head end shall provide suitable means for firmly holding the cap when placed over the writing head of the marker.

3.6.2 *Writing head.*—The writing head shall contain a spring-actuated valve and shall be of metal. The valve shall be self-aligning and constructed in such a manner as to prevent ink leakage when used in high temperatures (see 4.4.2). The head shall be threaded at the valve end to properly accommodate the internal threaded end of the body.

3.6.3 *Cap.*—The cap shall be of metal and shall entirely enclose the felt nib when assembled

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

to the marker. The cap shall be firmly held and shall not become loose when placed over the writing-head end of the body. A small hole shall be provided near the top of the cap so as to prevent the formation of a suction upon removal of the cap from the writing-head end.

3.6.4 *Class 1, small (pocket size).*—Class 1 markers shall be furnished with a felt nib and two replacement felt nibs of either of the following shapes: Conic or round, chisel or pointed, broad T-shape or square point. The end of the body opposite the writing head shall be suitable for holding the cap firmly when the marker is in use. The marker shall have an ink capacity of approximately ¼ ounce.

3.6.4.1 *Spring clip.*—A spring clip shall be permanently mounted on the cap with sufficient rigidity for supporting the marker when carried in the pocket. The spring clip shall be made of noncorrosive spring steel, plated spring steel, or plated phosphor bronze.

3.6.4.2 Class 1 markers shall conform to table I and shall be similar to figure 1.

3.6.5 *Class 2, large (heavy duty).*—Class 2 markers shall have an ink capacity of approximately ½ ounces or more. A removable hexagon screwcap at one end of the body, or other suitable means, shall be provided for access to the ink chamber. The marker shall conform to table II and shall be similar to figure 2.

TABLE I.—*Type I, class 1, valve control feed, small (pocket size) marker*

Length overall (including cap)		Body diameter		Weight (when empty)
Minimum	Maximum	Minimum	Maximum	Maximum
Inches	*Inches*	*Inch*	*Inch*	*Ounces*
5	6	0.450	0.625	1½

TABLE II.—*Type I, class 2, valve control feed, large (heavy duty) marker*

Length overall (including cap)		Body diameter		Weight (when empty)
Minimum	Maximum	Minimum	Maximum	Maximum
Inches	*Inches*	*Inch*	*Inch*	*Ounces*
7	8½	0.578	0.750	3

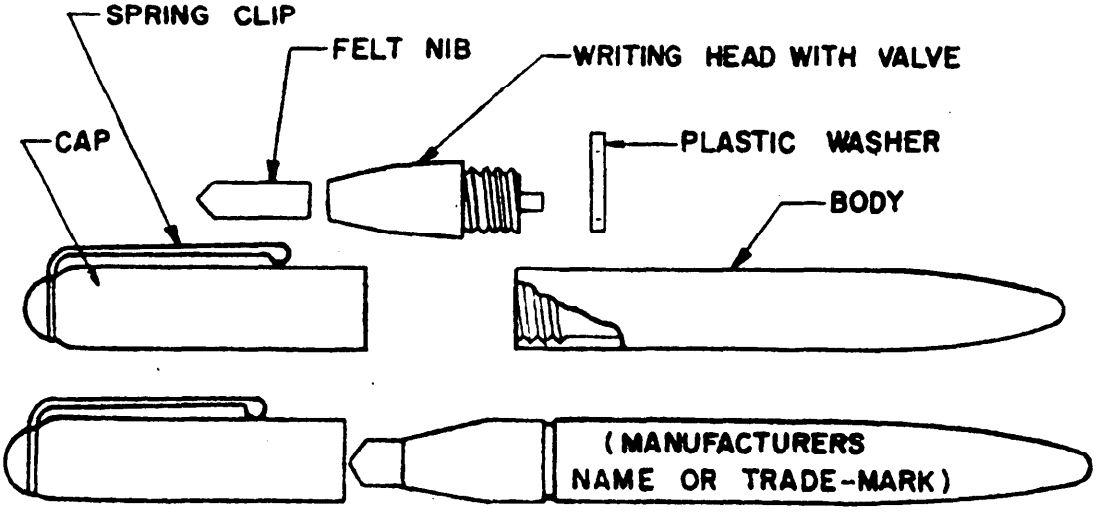


FIGURE 1.—Type I, class 1, valve control feed, small (pocket size) marker.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the currentversion before use.

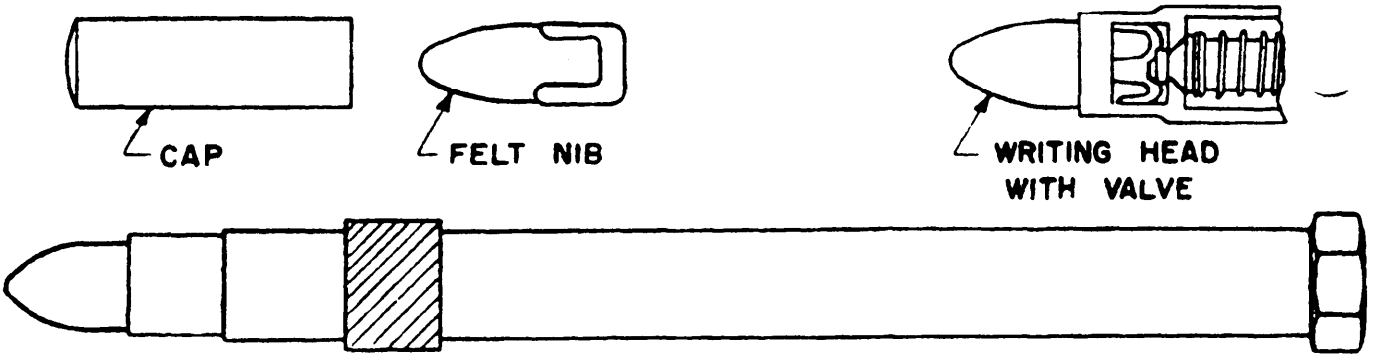


FIGURE 2.—Type I, class 2, valve control feed, large (heavy duty) marker.

3.7 Type II, adjustable control feed. — Type II markers shall be provided with an adjustable control valve to govern the ink flow, and shall consist essentially of a body or barrel, writing head, cap, and felt nib. The markers shall have an ink capacity of at least ½ ounce.

3.7.1 *Body (barrel).*—The body shall be in conformance with 3.6.1.

3.7.2 *Writing head.*—The writing head shall be of metal consisting of a valve, screwplug, and spring. The valve shall be self-aligning and constructed in such a manner as to prevent ink leakage when used in high temperatures (see 4.4.2). The flow of ink shall be governed by the spring tension applied on the valve by the screwplug. The screwplug when turned to the right or left shall regulate the spring tension, increasing or decreasing the flow of ink. The screwplug shall be provided with a screwdriver slot or other suitable means for adjusting the spring tension.

TABLE III.—*Type II, adjustable control feed marker*

Length overall (including cap)		Body diameter		Weight (when empty)
Minimum	Maximum	Minimum	Maximum	Maximum
Inches	*Inches*	*Inch*	*Inch*	*Ounces*
7¾	9¼	0.550	0.875	4

3.7.3 *Cap.*—The cap shall be in conformance with 3.6.3.

3.7.4 Type II markers shall conform to table III and shall be similar to figure 3.

3.8 Type III, gravity feed. — Type III markers shall consist essentially of a body or barrel, writing head, cap, spring clip, and felt nib.

3.8.1 *Class 1, small (pocket size).*

3.8.1.1 *Style A, metal.* — Style A markers shall conform to 3.6.3 through 3.6.4.1.

3.8.1.1.1 *Body (barrel).*—The body shall be of metal and shall be provided with internal threads at the open end of the ink chamber for attaching and securing the writing head. The outside diameter of the body at the writing-head end shall provide means for firmly holding the cap when placed over the writing head of the marker.

3.8.1.1.2 *Writing head.* — The writing head shall contain a wick or packing extending into the body which shall retain the ink and feed the felt nib by capillary action. The head shall be threaded at the wick or packing end to accommodate the internal threaded end of the body properly.

3.8.1.1.3 The markers shall be furnished in the sizes and conform to the dimensions shown in table IV, and shall be similar to figure 4. The size shall be as specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the currentversion before use.

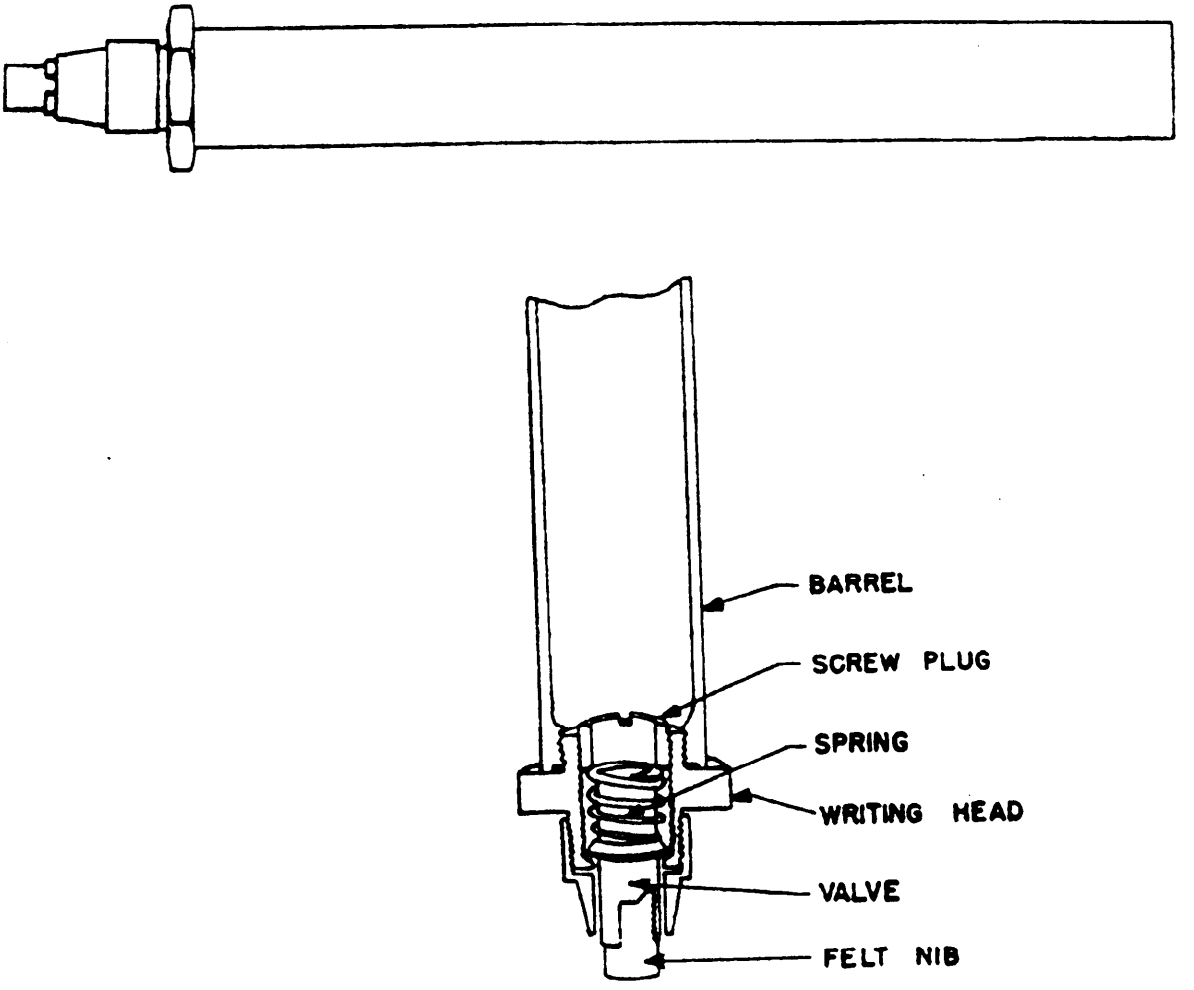


FIGURE 3.—Type II, adjustable control feed marker.

TABLE IV.—*Type III, class 1, style A, gravity feed, small (pocket size), metal marker*

Length overall (including cap)		Body diameter		Weight (when empty)
Minimum	Maximum	Minimum	Maximum	Maximum
Inches	*Inches*	*Inch*	*Inch*	*Ounces*
4⅝	4¾	0.475	0.525	⅞
5¼	5⅜	.525	.575	1⅝
7	7¼	.675	.750	2¾

3.8.1.2 *Style B, composition.* — Style B marker shall have an ink capacity of not less than ¼ ounce. The marker shall be of a composition such as phenolic resin or other suitable noninflammable material, except the writing head or collar containing the felt nib shall be of brass. The writing head or collar shall have an inside taper of a size to hold firmly to the tapered portion of the body to assure a leakproof connection when assembled. The marker shall be furnished with a felt nib and two replacement felt nibs, shall conform to table V, and shall be similar to figure 5.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the current version before use.

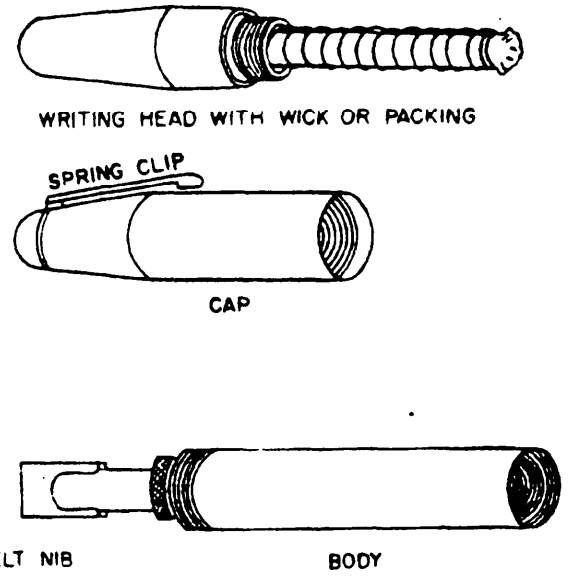


FIGURE 4.—Type III, class 1, style A, gravity feed, small (pocket size), metal marker.

3.8.1.3 *Style C, self-filling.*—Style C markers shall have a self-filling rubber bulb.

3.8.1.3.1 *Body (barrel).*—The body shall be of metal. The open end of the ink chamber at the writing-head end of the body shall have an inside taper of a size to hold firmly to the tapered portion of the writing head to assure a leakproof connection when assembled. The outside diameter of the body at the writing-head end shall provide suitable means for firmly holding the cap when placed over the writing head of the marker. The end of the body opposite the writing head shall be provided with means for accommodating and firmly holding the rubber bulb.

TABLE V.—*Type III, class 1, style B, gravity feed, small (pocket size), composition marker*

Length overall (including cap)		Body diameter		Weight (when empty)
Minimum	Maximum	Minimum	Maximum	Maximum
Inches	*Inches*	*Inch*	*Inch*	*Ounces*
5½	6½	0.450	0.550	1¼

3.8.1.3.2 *Writing head (collet).*—The writing head shall be of corrosion-resistant metal and consist of a split collet and a threaded draw nut. The collet shall be designed to permit easy adjustment of the felt nib and have a spring tension suitable for firmly gripping the nib when compressed by the draw nut. The connecting

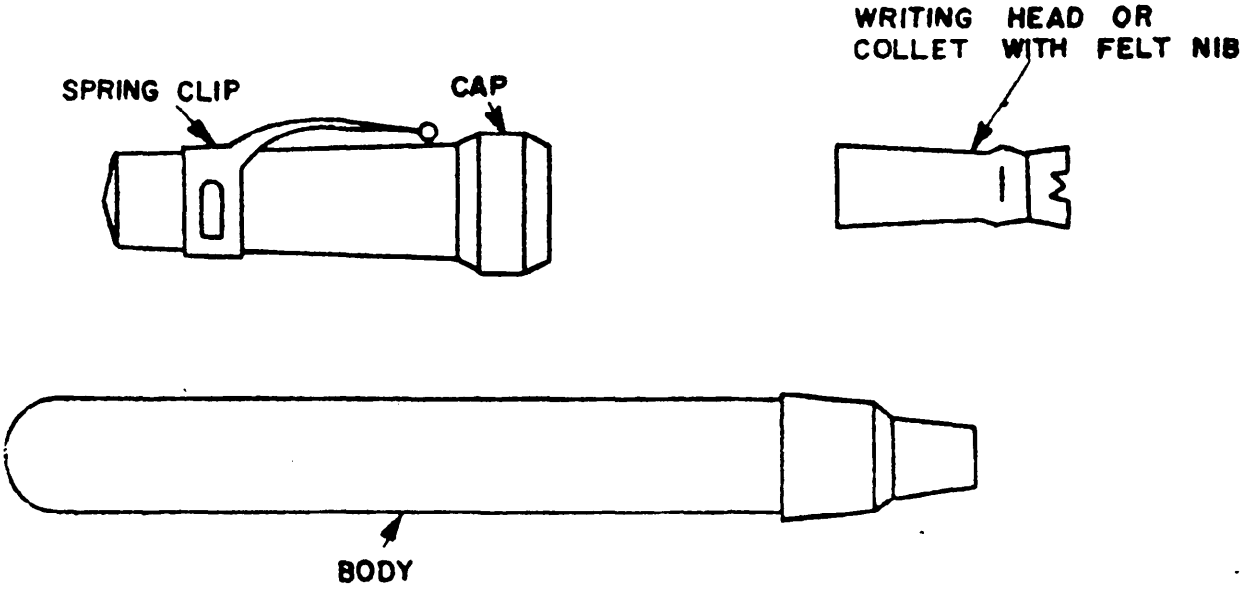


FIGURE 5.—Type III, class 1, style B, gravity feed, small (pocket size), composition marker.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE VI.—*Type III, class 1, style C, self-filling*

Diameter of nib	Length overall (including bulb)		Body diameter		Weight (when empty)	Capacity per filling (continuous uniform marking)
Nominal	Minimum	Maximum	Minimum	Maximum	Maximum	Minimum
Inch	*Inches*	*Inches*	*Inch*	*Inch*	*Ounces*	*Inches*
1/16	6 5/16	6 7/16	0.359	0.406	1 1/16	4,200
1/8	6 5/16	6 7/16	.359	.406	1 1/16	4,200
1/4	6 3/8	6 1/2	.359	.406	1 1/16	1,800
3/8	6	6 3/16	.484	.531	1 3/16	2,000

end of the writing head shall be tapered to correspond with the inside taper of the body.

3.8.1.3.3 *Felt nib.*—The felt nib shall be not less than 4 inches in length.

3.8.1.3.4 *Self-filling rubber bulb.*—The rubber bulb shall be in accordance with Military Specification MIL–R–2765. The bulb shall hold firmly to the body and be suitably designed for suction refilling of the marker. The bulb shall be constructed so as to conform to the test specified in 4.4.3.

3.8.1.3.5 *Spring clip.*—The spring clip shall be in conformance with 3.6.4.1 except it shall be mounted on the body.

3.8.1.3.6 *Cap.*—The cap shall be of metal, with the inside diameter shaped so as to conform to 3.8.1.3.1. The cap shall entirely enclose the felt nib and shall not become loose when assembled to the marker.

3.8.1.3.7 Style C markers shall conform to table VI and shall be similar to figure 6. The size shall be as specified (see 6.1).

3.9 Workmanship.—The workmanship shall be first class in every respect. Markers shall be free from imperfections or defects which may affect their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All markers of the same type, class, style, and size presented at one time shall be considered a lot for the purposes of inspection and tests.

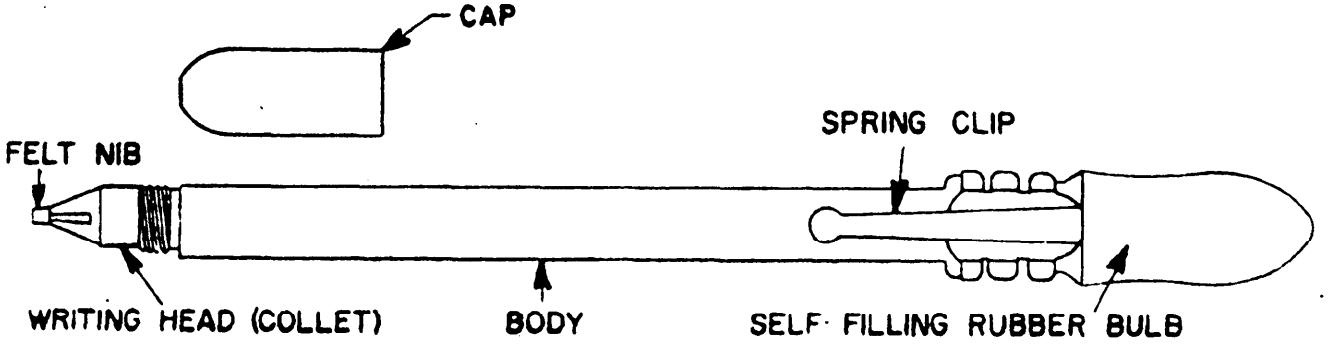


FIGURE 6.—Type III, class 1, style C, self-filling.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the current version before use.

4.1.2 *Sampling for visual and dimensional inspection.*—A random sample of markers shall be taken by the Government inspector in accordance with table VII from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with Military Standard MIL–STD–105.

TABLE VII.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective*

Number of markers in lot	Number of markers in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110 ...	15	0	1
111 to 300 ...	25	1	2
301 to 500 ...	35	1	2
501 to 800 ...	50	2	3
801 to 1,300 ..	75	3	4
1,301 and over .	110	4	5

NOTE.—The Government inspector may institute tightened or reduced inspection in accordance with Section 9 of Military Standard MIL–STD–105.

4.1.3 *Sampling for tests.*—A random sample of markers shall be taken by the Government inspector from each inspection lot in accordance with table VIII and shall be subjected to the tests specified in 4.3.

TABLE VIII.—*Sampling for tests AQL (approx.) = 1.0 percent defective*

Number of markers in lot	Number of markers in sample	Acceptance number (defective)	Rejection number (defective)
15 and under	5	0	1
16 to 40	7	0	1
41 to 110 ...	10	0	1
111 to 300 ...	15	0	1
301 to 500 ...	25	0	1
501 to 800 ...	35	1	2
801 to 1,300 ..	50	1	2
1,301 and over .	75	2	3

4.2 Inspection.

4.2.1 *Visual and dimensional inspection.*—Each of the sample markers selected in accordance with table VII shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any marker in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective markers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Tests.—Each of the sample markers selected in accordance with 4.1.3 shall be subjected to the tests specified in 4.4.

4.4 Test procedures.

4.4.1 *Performance test.*—Types I, II, and III sample markers shall be subjected to a short practical test by the Government inspector when marking with the marker. The Government inspector shall observe the balance in affording ease of handling and legibility when marking with a slight downward pressure. No leakage shall appear when marker is held in any position, except type III, style C marker shall not be held by the self-filling rubber bulb. The cap shall be removed and replaced with ease, and shall hold firmly when attached to body.

4.4.2 *Temperature test.*—Types I and II markers shall be subjected to a temperature test specified hereinafter to determine the flow of ink relative to the operation of the valve (see 3.6.2 and 3.7.2). The markers shall be tested using a thin dye ink and an increased temperature. A receptacle, chest, or other suitable means shall be provided for regulating temperatures of from 100° to 130° Fahrenheit (F.).

4.4.2.1 *Increased temperature test.*—The marker shall be furnished with a new felt nib and shall be filled with a thin dye ink. In room temperatures of 70° to 78°F., the marker shall be placed in a testing chamber or receptacle with a temperature of 100°F. for 10 minutes. In room temperatures of 78° to 90°F., the temperature

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the currentversion before use.

of the testing chamber or receptacle shall be increased to 120°F. When the marker is removed from the testing chamber or receptacle, examination of the felt nib shall show no indication of saturation or discoloration due to leakage of the valve. If the felt nib indicates that ink leakage of the valve has occurred due to increased pressure created in the body, the marker shall be rejected.

4.4.2.2 *Thin dye ink.*—The ink used in the temperature test shall be thin, fast-flowing dye ink, red or black in color. The ink shall be quick-drying and waterproof, and shall consist essentially of a fast dye dissolved in an organic solvent. The ink shall not have a deleterious effect on the markers and shall not exhibit coagulation, granulation or precipitation under ordinary working or storage conditions. The ink shall not contain benzol (benzene), aniline oil, chlorinated compounds, or other toxic or hydrolyzable chlorine derivatives.

4.4.3 *Rubber bulb performance test.*—Type III, style C markers shall be subjected to a test to determine the performance of the self-filling rubber bulb. The marker to be tested shall be inserted into a vessel containing thin dye ink (see 4.4.2.2) and filled by repeatedly depressing the rubber bulb a sufficient number of times to insure complete saturation of the nib. The marker shall then be removed from the ink and by slowly depressing the rubber bulb, eject the excess ink. The marker shall be reinserted into the ink and the bulb depressed once. When the bulb has regained its natural shape the marker shall be removed from the ink and by slowly depressing the rubber bulb, again eject excess ink. This process shall be repeated 30 times. If the marker fails to eject any excess ink after the initial saturation of the nib then the marker does not comply with the rubber bulb performance test.

4.4.4 *Nib capacity test.*—Type III, style C markers shall be subjected to a test to determine compliance with the marking capacity requirements of table VI. The marker to be tested shall be furnished with a new felt nib and, after being properly filled, shall show continuous uniform marking complying with the requirements of table VI on newsprint paper without a noticeable change in color intensity.

4.5 **Rejected lots.**—Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming markers. The Government inspector shall again take at random and examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Civil agencies.

5.1.1 *Packaging.*—Markers shall be cleaned, preserved, and packaged in accordance with the manufacturer's commercial practice.

5.1.2 *Packing.*—Markers shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Containers shall comply with the Consolidated Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.1.3. *Marking.*

5.1.3.1 *Issue packages.* — Unless otherwise specified, packages shall be marked with the name and type, class, and style of material, the quantity contained therein, and the name of the contractor.

5.1.3.2 *Shipping containers.* — Unless otherwise specified, containers shall be marked with the name and type, class, and style of material, the quantity of each type contained therein, name of the contractor, number of the contract or order, name and address of the consignee, and the gross weight.

5.2 Military agencies.

5.2.1 *Preparation for shipment.* — Cleaning, preservation, packaging, packing, and marking

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the currentversion before use.

for shipment shall be in accordance with Military Specification MIL-P-15424, except that plated, enameled, or anodized external surfaces will not require preservative compound or wrapping as specified therein.

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type, class, and style of marker required (see 1.2).

(c) Type, shape, and quantity of felt nibs if additional nibs are desired (see 3.3.1, 3.3.2, and 3.3.3).

(d) Size of type III, class 1, style A or C required (see 3.8.1.1.3 and 3.8.1.3.7).

(e) Whether preparation for domestic shipment, immediate use, domestic shipment and storage, or overseas shipment is required.

Purchasers should also exercise any desired options offered herein. (See 5.1.3.1 and 5.1.3.2.)

6.1.1 When ordering replacement nibs, the following should be specified:

(a) Manufacturer's name or trade-mark of marking pen for which the nibs are required.

(b) Type, shape, and quantity of felt nibs (see 3.3.1, 3.3.2, and 3.3.3).

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—Q E O

Navy—Sh MC O.

Air Force.

Copies of this specification may be purchased for 15 cents each.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:48Z
Check the source to verify that this is the current version before use.